

Jonathan Liu

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EDUCATION

University of Washington

Seattle, WA

Master of Science in Electrical and Computer Engineering

Sep 2027 – June 2028

Bachelor of Science in Electrical and Computer Engineering | GPA: 3.93

Sep 2024 – June 2027

- **Honors:** Combined BS/MS (Analog and Mixed-Signal Design), Frey Endowed EE Scholarship (2026–27), IEEE HKN Honor Society (Top 8% ECE), Tau Beta Pi Engineering Honor Society (Top 12.5%), Annual Dean's List (2024–2026).
- **Relevant Coursework:** Analog IC Design, Microelectronic Circuits, Digital Circuit Design, Signals & Systems, Computer Architecture.

UPWARDS Semiconductor Intensive – Hiroshima University

Hiroshima, Japan

CMOS Design–Fabrication–Test Practical Training | GPA: 4.0

July – Aug 2026

- Selected for the **UPWARDS** U.S.–Japan consortium with **Micron Technology** and **Tokyo Electron** (approved leave from Trellis).
- Designed a **CMOS inverter layout** in **Siemens L-Edit**, ran the photolithography-to-metal cleanroom flow, and performed wafer-probe **I–V characterization** on a **Keysight B1500A**, comparing measured against simulated switching.

RELEVANT EXPERIENCE

Hardware Engineering Intern

June – Sep 2026

Trellis Robotics

Mountain View, CA

- Designed a **4-layer, 11-IC robot motor-control and battery PCB** in KiCad (dual motor drivers, 12 V/5 V/3.3 V rails, 4S battery management), owning schematic, layout, DFM, and JLCPCB fabrication end to end.
- Diagnosed a ground-split failure that **destroyed two first-revision boards** and re-architected a 4S battery-management system around a **high-side TI BQ76922**, eliminating the failure mode.
- Brought up a redesigned board **the day it arrived with zero hardware rework**, then fixed wake-from-shutdown and added a hardware power switch through on-chip OTP programming, **avoiding a board respin**.
- Developed **20,000+ lines of C firmware** from scratch for an ESP32-P4 robot controller, including I²C drivers for 6 peripheral ICs, a watchdog motor failsafe, and **1080p video at 27 fps**.
- Engineered a **GPS-free 3D pipe-localization system** from only an IMU and a tether encoder, syncing two independent clocks and detecting pipe bends statistically; achieved **4.3 cm RMS accuracy** over 19 runs.
- Integrated LLMs into engineering: shipped a **source-grounded AI assistant** that answers from 6,700+ code and doc chunks with cross-model fact checking, and open-sourced agentic PCB review tooling.

Hardware Lead

Oct 2024 – Present

Husky Robotics Drone Subteam

Seattle, WA

- Led a 20+ person EE team and managed three custom PCBs in parallel: a **flight controller**, an electronic speed controller, and a power distribution board, including design reviews and the power and signal interfaces between them.
- Designed a 4-layer **STM32H743** flight controller in **Altium**: schematic, layout, 5 V/3.3 V regulation, current sensing, and the connector pinout for motors, radios, and sensors.
- Led electrical **system integration** for a University Rover Challenge quadrotor: selected and wired a 6S Li-Ion pack, 750 kV motors, 50 A ESCs, Pixhawk 6C, ELRS radios, HDZero video, and a Jetson Orin Nano, including power split, harness, and **MAVLink**.

PROJECTS

Single-Stage 5T OTA in 45nm CMOS (GPDK045)

May 2026

- Designed a single-stage **5T OTA** in **Cadence Virtuoso** on **GPDK045** (45nm), hand-sizing the input pair, current-mirror load, and tail source from **gm/I_D** analysis.
- Verified the design in **ADE Explorer/Assembler** across AC, DC, and transient sweeps: **1.20 GHz** gain-bandwidth, **1 V** peak-to-peak swing at 100 MHz on **5 mW**.

Adjustable AC-DC Boost Converter

June 2026

- Designed, simulated in **LTspice**, and **breadboarded** a two-stage **AC-DC boost** converter delivering an adjustable **+10–21 VDC** rail at **0.4%** load regulation under **100 mV** ripple.

FPGARhythm

March 2026

- Developed a hardware-based rhythm game in **SystemVerilog** on an Intel **Cyclone V** FPGA (DE1-SoC), processing user inputs and rendering cascading notes via **VGA** output.

Three-Band Active Audio Equalizer

March 2026

- Designed and built a three-band active equalizer around the **LM348** op-amp; derived the transfer functions by hand and validated them in **LTspice**.

Autonomous Line-Following Car

Feb 2026

- Built an Arduino line-follower with a tuned **PID** controller; **all-time fastest record** in the UW Line-Following Car competition.

SKILLS

IC Design & Test: Cadence Virtuoso (schematic, layout, extraction), ADE Explorer/Assembler, GPDK045 45nm CMOS, Siemens L-Edit, LTspice/SPIICE, gm/I_D sizing, cleanroom CMOS fabrication, Keysight B1500A wafer probing, I–V characterization.

PCB & Embedded: Altium Designer, KiCad, multi-layer layout, multi-rail PDN, power sequencing, IPC-2221/DFM, STM32 (CubeIDE/HAL), ESP-IDF, Arduino, UART/I²C/SPI/CAN, battery management, motor drivers, buck converters, LDOs.

Digital & Software: SystemVerilog, FPGA (Quartus, ModelSim), C/C++, Python (NumPy/SciPy), MATLAB, Linux/Bash, Git.

Languages: English, Chinese (Mandarin & Cantonese), Japanese.

Leadership: President, IEEE UW Student Chapter; **Events Officer**, Hong Kong Student Association at UW.